Stat 256: Causality

Instrumental Variables

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Outline

1. IV setup
2. The new way: with potential outcomes and heterogeneity
   - Identification
   - Estimation
3. What about ID under DAGs?
4. Sensitivity for IV (OVB framework)

Where are we? Where are we going?

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- First up: Instrumental variables (IV): "what if you can find anexogenous sources of variation that drives the treatment?"

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- But it isn't the effect of the program. How do we get back to that?

Backing it out: four kinds of people

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Punchline: the ITT over everyone will be an average over these and so will be a *watered-down* version of the program effect. Specifically,

Backing it out: the algebra

Averaging the ITT over the four types:

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\text{ITT} &= \underbrace{(\text{ITT} \mid \text{never-taker})}_{=0} \cdot \Pr(\text{never-taker}) \\
&+ \underbrace{(\text{ITT} \mid \text{always-taker})}_{=0} \cdot \Pr(\text{always-taker}) \\
&+ (\text{ITT} \mid \text{complier}) \cdot \Pr(\text{complier}) \\
&+ \underbrace{(\text{ITT} \mid \text{defier})}_{=0,\ \text{by assumption}} \cdot \Pr(\text{defier})
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And for compliers, the ITT is the ATE.

From ITT-on-compliers to LATE

One more thing you need to know: $\Pr(\text{complier})$ is the "first stage" effect:

$\Pr(\text{complier}) = \mathbb{E}[D_1 - D_0] = \mathbb{E}[D|Z = 1] - \mathbb{E}[D|Z = 0]$ (by ignorability of Z), which equals the proportion of compliers (if no defiers).

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Putting it together: We call the ATE among the compliers the "local average treatment effect" (LATE):

$$\text{LATE} = \frac{\text{ITT}}{\Pr(\text{complier})} = \frac{\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0]}{\mathbb{E}[D \mid Z = 1] - \mathbb{E}[D \mid Z = 0]}$$

Basic IV setup with DAGs

$$\begin{tikzcd} & U \arrow[dl, dashed] \arrow[dr, dashed] & \\ Z \arrow[r] & D \arrow[r] & Y \end{tikzcd}$$

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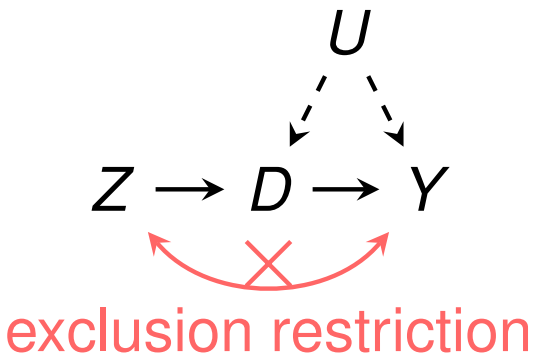

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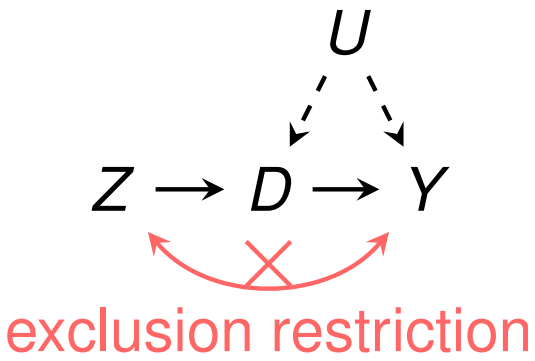

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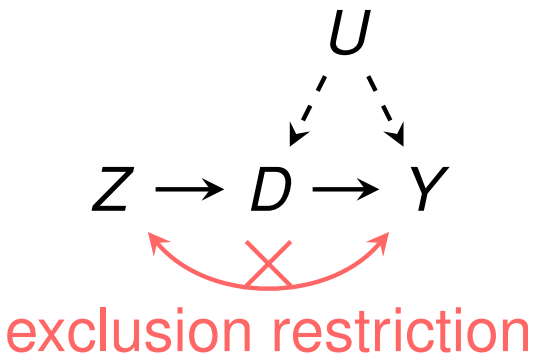

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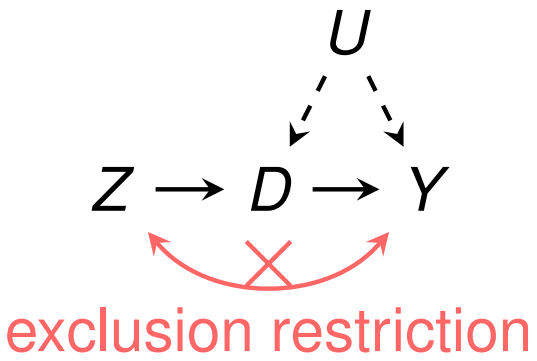

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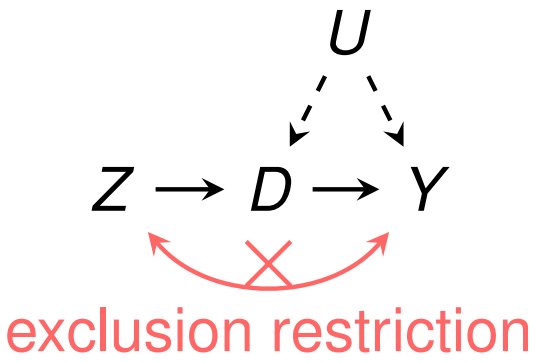

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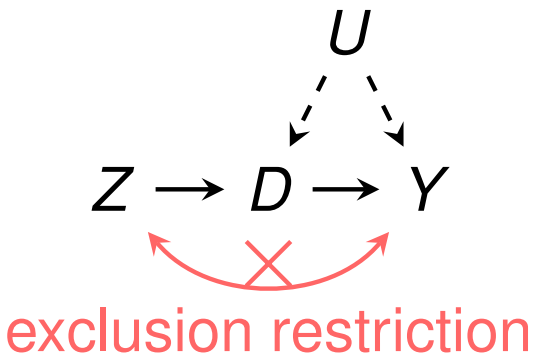

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- Actually Z-D confounding is workable, but ruled out in one popular framework we will look at.

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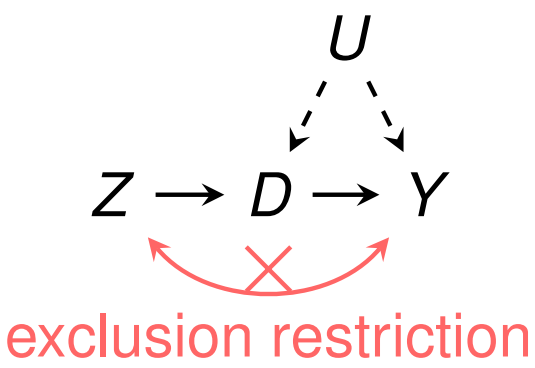

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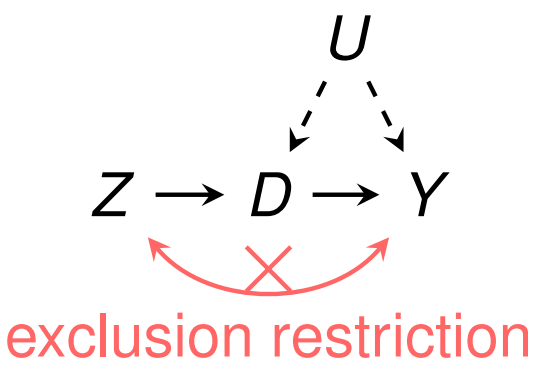

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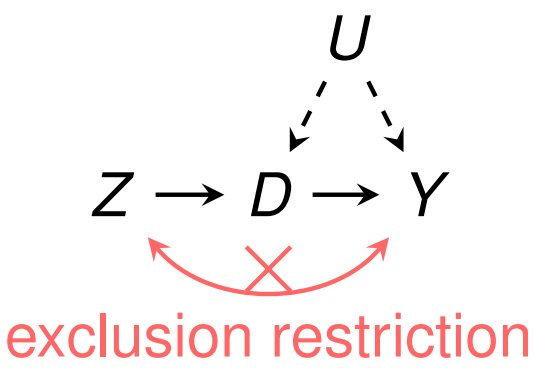

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- But, DAGs+do-calc won't tell you that IV identifies a causal effect:
  - Identification depends on a "monotonicity" assumption (Z only affects D in one direction)

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Uses of IV with randomization:

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- Acharya, Blackwell, Sen (2015): cotton suitability as IV for proportion of population in slavery in 1860 (outcome is white attitudes today)

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- We'll work mostly in the modern *potential outcomes* language — LATE, compliers, monotonicity — which makes assumptions and target estimand transparent under heterogeneity.
- The SEM/2SLS treatment is in the appendix for reference (and because you'll see it everywhere in applied work).

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POM for Instrumental Variables

Definition (Instrument)

Z_i: Binary instrument for *unit i*.

$$Z_i = \begin{cases} 1 & \text{if unit } i \text{ “encouraged” to receive treatment} \\ 0 & \text{if unit } i \text{ “encouraged” to receive control} \end{cases}$$

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Assumption (Consistency of potential treatments)

Observed treatments are realized as

$$D_i = Z_i \cdot D_{1i} + (1 - Z_i) \cdot D_{0i} \text{ so } D_i = \begin{cases} D_{1i} & \text{if } Z_i = 1 \\ D_{0i} & \text{if } Z_i = 0 \end{cases}$$

Potential Outcome Model for Instrumental Variables

Following Angrist, Imbens, and Rubin (1996), we can define:

Definition

- *Compliers*: $D_{1i} > D_{0i}$ ($D_{0i} = 0$ and $D_{1i} = 1$).
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- Though, further assumptions often help, e.g. no defiers, and sometimes no always-takers.

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Definition (Potential Outcomes)

Given the binary instrument $Z_i \in \{1, 0\}$ and the binary treatment $D_i \in \{1, 0\}$ every unit now has four potential outcomes $Y_i(D, Z)$:

$$Y_i(D_i = 1, Z_i = 1);\ Y_i(D_i = 1, Z_i = 0);\ Y_i(D_i = 0, Z_i = 1);\ Y(D_i = 0, Z_i = 0)$$

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- Under exclusion restriction, we will have $Y_i(d, z = 0) = Y(d, z = 1) = Y_i(d)$

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NB: this estimand varies with the particular instrument Z.

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Some special cases:

- When the treatment intake, D_i, is itself randomized, then $Z_i = D_i$ and every individual is a complier
- If one-sided noncompliance s.t. $D_{0i} = 0$:

$$\mathbb{E}[Y_{1i} | D_{1i} > D_{0i}] = \mathbb{E}[Y_{1i} | Z_i = 1, D_{1i} = 1] = \mathbb{E}[Y_{1i} | D_i = 1]$$

And

$$\mathbb{E}[Y_{0i} | D_{1i} > D_{0i}] = \mathbb{E}[Y_{0i} | D_i = 1]$$

Thus,

$$\alpha_{LATE} = \mathbb{E}[Y_{1i} - Y_{0i} | D_{1i} > D_{0i}] = \mathbb{E}[Y_{1i} - Y_{0i} | D_i = 1] = \alpha_{ATT}$$

Outline

1. IV setup
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   - Identification
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Identification of the LATE under IV

Identification Assumption

1. *Ignorability of the Instrument:* $(Y_{0i}, Y_{1i}, D_{0i}, D_{1i}) \perp\!\!\!\perp Z_i$
2. *First Stage:* $P(D_{1i} = 1) \neq P(D_{0i} = 1)$
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Identification Result

$$\begin{aligned}\mathbb{E}[Y_1 - Y_0 | D_1 > D_0] &= \frac{\mathbb{E}[Y|Z=1] - \mathbb{E}[Y|Z=0]}{\mathbb{E}[D|Z=1] - \mathbb{E}[D|Z=0]} \\ &= \frac{\textit{Intent to Treat Effect of } Z \textit{ on } Y}{\textit{First Stage Effect of } Z \textit{ on } D} \\ &= \frac{\textit{Intent to Treat Effect}}{\textit{Proportion of Compliers}}\end{aligned}$$

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$$ITT = ATE(compliers)Pr(complier) + 0 + 0$$

$$ATE(compliers) = \frac{ITT}{Pr(compliers)}$$

Identification with Instrumental Variables

Identification Assumption

1. *Ignorability of the Instrument:* $(Y_0, Y_1, D_0, D_1) \perp\!\!\!\perp Z$
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Proof.

$$\frac{E[Y|Z=1] - E[Y|Z=0]}{E[D|Z=1] - E[D|Z=0]} = \frac{E[Y_0 + (Y_1 - Y_0)D_1|Z=1] - E[Y_0 + (Y_1 - Y_0)D_0|Z=0]}{E[D_1|Z=1] - E[D_0|Z=0]}$$

$$= \frac{E[Y_0 + (Y_1 - Y_0)D_1] - E[Y_0 + (Y_1 - Y_0)D_0]}{E[D_1] - E[D_0]} = \frac{E[(Y_1 - Y_0)(D_1 - D_0)]}{E[D_1 - D_0]}$$

$$= \frac{E[Y_1 - Y_0|D_1 > D_0]P(D_1 > D_0) - E[Y_1 - Y_0|D_1 < D_0]P(D_1 < D_0)}{E[D_1 - D_0]} \text{ as } (D_1 - D_0)\text{=}(1, 0, -1)$$

$$= \frac{E[Y_1 - Y_0|D_1 > D_0]P[D_1 > D_0]}{P(D_1 > D_0)} = E[Y_1 - Y_0|D_1 > D_0]$$

□

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Monotonicity: $D_1 \geq D_0$

- Rules out defiers. Not testable but sometimes qualitatively easy to argue.

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Standard implementation is the IV estimator:

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3. However, with heterogenous effects, β_{IV} estimated in this way does not necessarily have a clear causal interpretation!
   - complier group changes depending on strata of X.
   - κ-weighting: weights s.t. any model you run gets you a result as if only estimated for compliers (Abadie, 2003): Local average response function (LARF).

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DAGs were useful in talking through IV, but unsatisfying that they do not identify IV.

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One proposal[1]: Sub-population DAGs for compliers and non-compliers (rule-out defiers),

- for compliers, the SCM says $D = f_D(Z) = Z$.
- for non-compliers, $D = f_D(U_D)$.

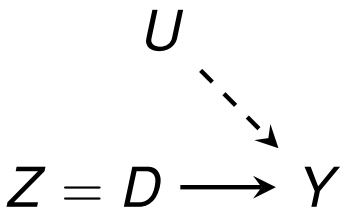

Figure: Among compliers

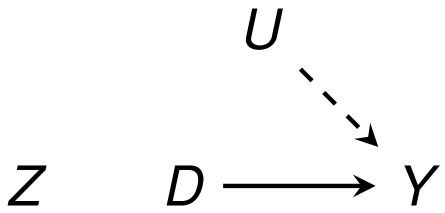

Figure: Among noncompliers

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IV identification in DAGs

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Further,

- Because $D = Z$ for compliers, $p(y|do(z)) = p(y|do(d))$ for compliers.
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$$\begin{aligned} p(y|do(z)) &= p(y|do(z), \text{comp=1})p(\text{comp=1}) + p(y|do(z), \text{comp=0})p(\text{comp=0}) \\ &= p(y|do(d), \text{comp=1})p(\text{comp=1}) + p(y|do(d), \text{comp=0})p(\text{comp=0}) \end{aligned}$$

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Thus for $ITT = p(y|do(z=1)) - p(y|do(z=0))$ we have

$$\begin{aligned} &= [p(y|do(d=1), \text{comp=1})p(\text{comp}) + p(y|do(d=1), \text{comp=0})p(\text{comp=0})] - \\ &\quad [p(y|do(d=0), \text{comp=1})p(\text{comp}) + p(y|do(d=0), \text{comp=0})p(\text{comp=0})] \\ &= [p(y|do(d=1), \text{comp=1}) - p(y|do(d=0), \text{comp=1})]p(\text{comp}) \\ &= LATE \cdot p(\text{comp=1}) \end{aligned}$$

Summary

DAGs communicate the IV setup well, but

- We needed to add in non-NP assumptions. Here, it is that population is split into those with different $D = f_D(U)$ and those with $D = f_D(Z, U) = Z$.
- There is nobody for whom $D = f_D(Z, U) = 1 - Z$ (monotonicity)
- U is the cause of compliance (or not); the partition of units into compliers vs. non-compliers relates to a partition of U.
- It would be interesting to ponder what else can be handled by such population-splitting tricks.

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Back to the big picture:

- IV was our first strategy outside of do-calculus
- It is very widely used in empirical practice.
- We will talk about sensitivity analysis for IV
- But first we will look at other non-do-calc-identifiable opportunities, i.e. those that flow from imposing other "non-NP" type assumptions.

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OVB with instrumental variables

We can apply the same OVB machinery to IV. Walks through, applied to the Card (1993) returns-to-education example.

Running example: returns to education (Card 1993)

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- Confounding may be severe: ability, motivation, family/social.
- Card uses *proximity to college* as instrument.

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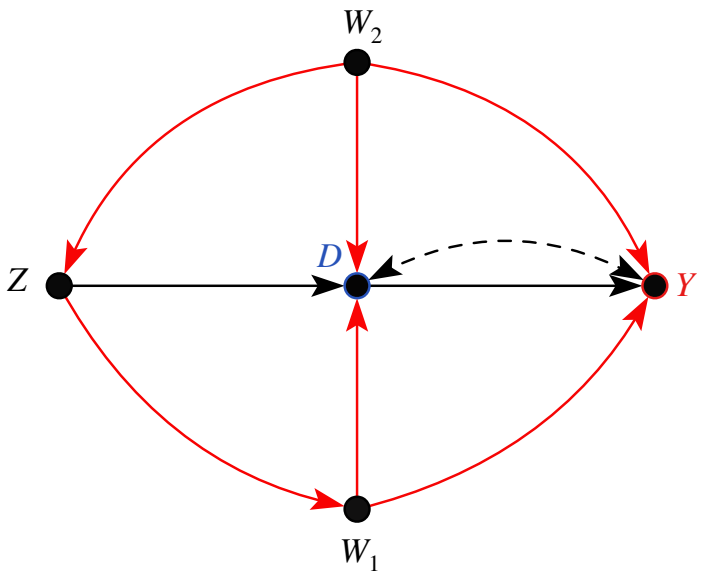

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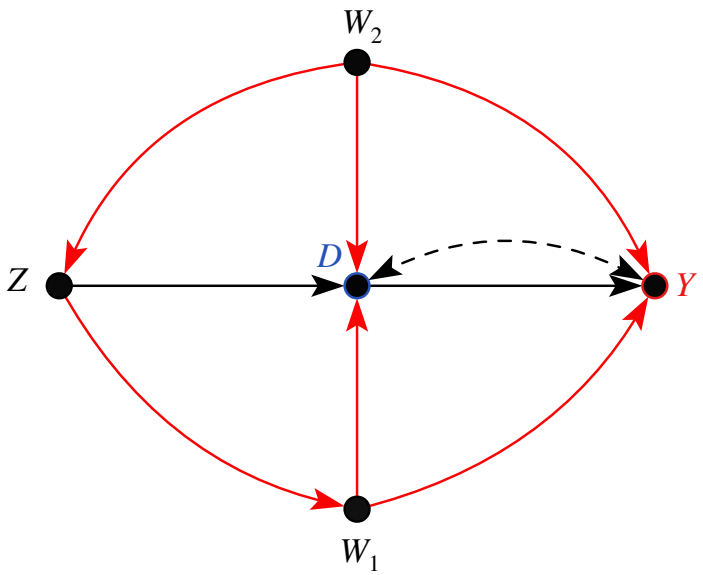

How "bad" would such violations need to be to change the conclusion?

IV regression: three estimators

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$$\text{Reduced form:} \quad \text{Earnings} = \hat{\lambda}_{\text{res}}\,\text{Proximity} + \mathbf{X}\hat{\beta}_{\text{res}} + \hat{\epsilon}_{y,\text{res}}$$

$$\text{First stage:} \quad \text{Education} = \hat{\theta}_{\text{res}}\,\text{Proximity} + \mathbf{X}\hat{\psi}_{\text{res}} + \hat{\epsilon}_{d,\text{res}}$$

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Same answer:

$$\hat{\tau}_{\text{2SLS}} = \hat{\tau}_{\text{ILS}} := \frac{\hat{\lambda}_{\text{res}}}{\hat{\theta}_{\text{res}}} = \frac{0.042}{0.319} \approx 0.132.$$

IV regression: Anderson–Rubin

3. Anderson–Rubin (Fieller):

$$\text{Earnings} - \tau_0 \text{Education} = \hat{\phi}_{\tau_0,\text{res}} \text{Proximity} + \mathbf{X}\hat{\beta}_{\tau_0,\text{res}} + \hat{\epsilon}_{\tau_0,\text{res}}$$

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How would these change had we included W?

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But W was not observed, so we ran

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Start with the reduced form

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How would including W change inference about λ?

Same OVB analysis, different variable names

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&= \text{SeF} \sqrt{\text{df} \, / (\text{df} - 1)} \cdot \widehat{\text{se}}(\hat{\lambda}_{\text{res}})
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$$t^{\dagger}_{\alpha,\text{df}-1,\boldsymbol{R}^2} := \text{SeF}\sqrt{\text{df}/(\text{df}-1)} \cdot t^*_{\alpha,\text{df}-1} + \text{BF}\sqrt{\text{df}}.$$

Routine reporting: (extreme) robustness values

What would it take to reduce the coefficient until we no longer reject (with confidence $1 - \alpha$) that it is $100q\%$ as large?

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Using the RF coefficient:

- $\text{RV}_{q,\alpha}(\lambda)$: minimal strength of association with both Z and Y needed.
- $\text{XRV}_{q,\alpha}(\lambda)$: minimal strength with Z alone, even if W explains *all* residual outcome variance.

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Sensitivity of first stage: If the sign of the FS isn't guaranteed, you cannot rule out unbounded estimates / CIs.
AR is consistent with this logic.

Sensitivity with the AR regression

$$Y_{\tau_0} = \hat{\phi}_{\tau_0,\text{res}} Z + \mathbf{X}\hat{\beta}_{\tau_0,\text{res}} + \hat{\epsilon}_{\tau_0,\text{res}}$$

The AR confidence interval:

$$\text{CI}_{1-\alpha}(\tau) = \left\{ \tau_0 : \; t^2_{\hat{\phi}_{\text{res},\tau_0}} \leq (t^*_{\alpha,\text{df}-1})^2 \right\}.$$

Replace $t^*_{\alpha,\text{df}-1}$ with $t^{\dagger\ \max}_{\alpha,\text{df}-1,\boldsymbol{R}^2}$:

$$\text{CI}^{\max}_{1-\alpha,\boldsymbol{R}^2}(\tau) = \left\{ \tau_0 : \; t^2_{\hat{\phi}_{\text{res},\tau_0}} \leq (t^{\dagger\ \max}_{\alpha,\text{df}-1,\boldsymbol{R}^2})^2 \right\}.$$

Analytical solution via quadratic inequality. Yields $\text{RV}_{q,\alpha}(\tau)$, $\text{XRV}_{q,\alpha}(\tau)$ and $\text{RV}_{\geq q,\alpha}(\tau)$, $\text{XRV}_{\geq q,\alpha}(\tau)$.

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1. Reduced form: $H : \tau = 0$**.**

Outcome: *Earnings* (log)

Instrument	Est.	SE	t	$R^2_{Y\sim Z\mid \mathbf{X}}$	$\text{XRV}_{q^*,\alpha}$	$\text{RV}_{q^*,\alpha}$
Proximity	0.042	0.018	2.33	0.18%	0.05%	0.67%

Bound (1× smsa): $R^2_{Y\sim W\mid Z,\mathbf{X}} = 2\%$, $R^2_{Z\sim W\mid \mathbf{X}} = 0.6\%$, $t^\dagger = 2.55$
$\text{df} = 2994,\ q^* = 1,\ \alpha = 0.05$

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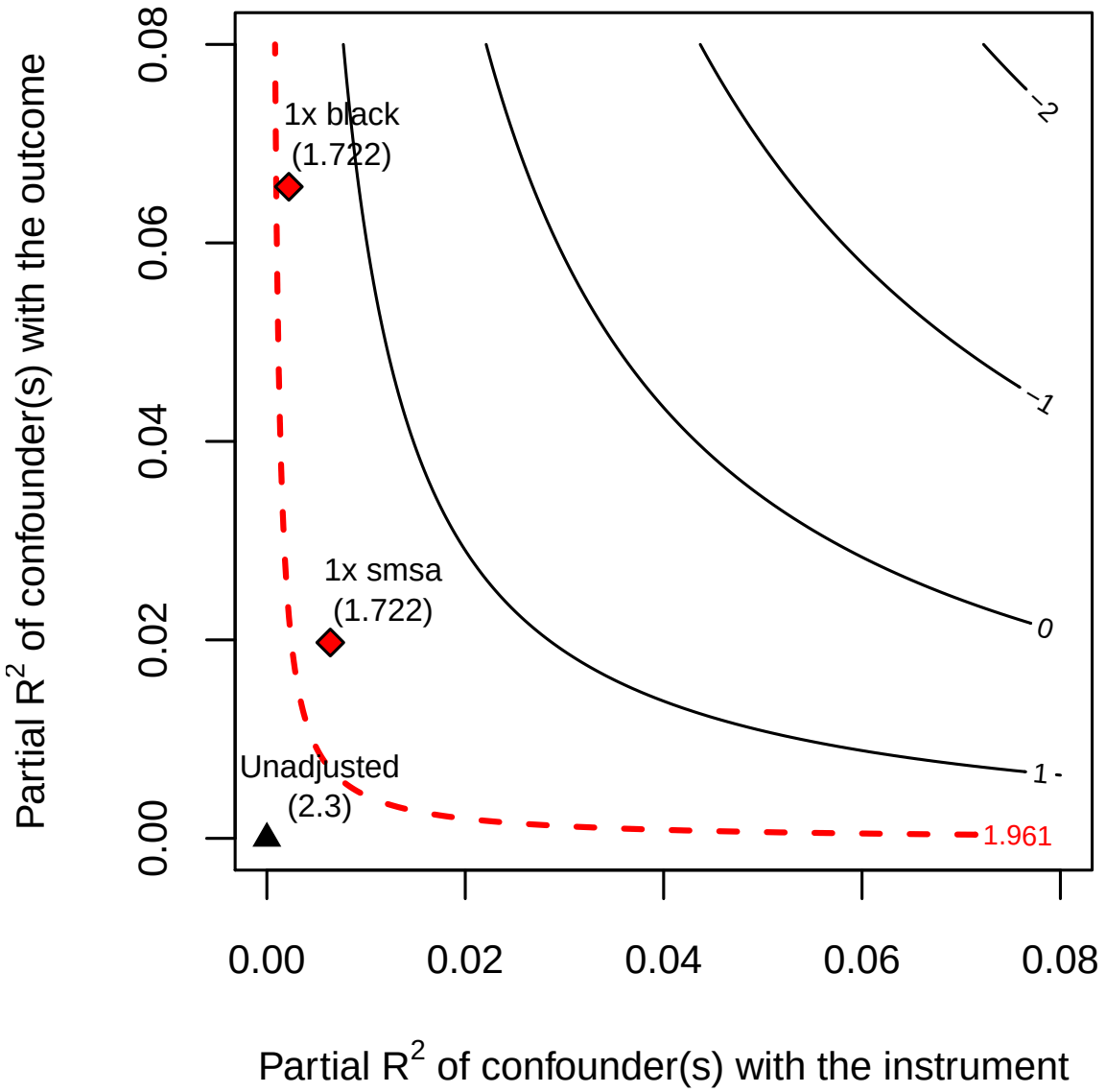
2. First stage: stability.

Outcome: *Education* (years)

Instrument	Est.	SE	t	$R^2_{D\sim Z\mid \mathbf{X}}$	$\text{XRV}_{q^*,\alpha}$	$\text{RV}_{q^*,\alpha}$
Proximity	0.32	0.088	3.64	0.44%	0.31%	3.02%
Bound (1× smsa): $R^2_{D\sim W\mid Z,\mathbf{X}} = 0.5\%$, $R^2_{Z\sim W\mid \mathbf{X}} = 0.6\%$, $t^\dagger = 2.26$						

Card example: reduced form contours



Card example: sensitivity of the IV (AR)

Outcome: *Earnings* (log)

Treatment	Est.	LL	UL	t	$\text{XRV}_{\geq q^*,\alpha}$	$\text{RV}_{\geq q^*,\alpha}$
Education	0.132	0.025	0.285	2.33	0.05%	0.67%

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- Identical to RF results because the null is $\tau = 0$.
- More interesting under other null hypotheses.

Card example: CI-limit contours

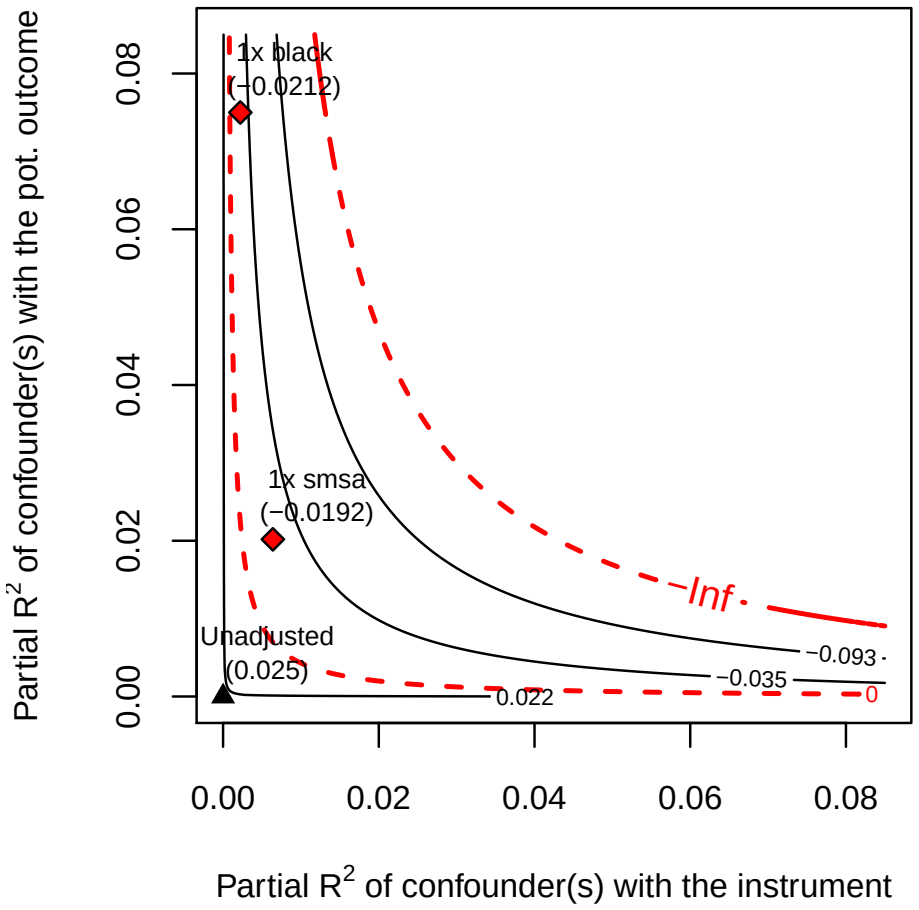

Figure: Sensitivity contours: lower 95% limit

IV sensitivity: conclusions

- OLS sensitivity to confounding can be as simple as adjusting your critical value.
- For null of zero in IV, sensitivity of the reduced form *is all you need*.
- General sensitivity for IV is also easy: in the AR framework, explore how postulated confounding moves point estimates, CI limits, RVs, and bounds.

Outline

5 Appendix: SEM / 2SLS framing of IV

IV with constant effects

- Write down a causal model for Y_i with constant effects and an unmeasured confounder, U_i:

$$Y_i(d, u) = \alpha + \tau d + \gamma u + \eta_i$$

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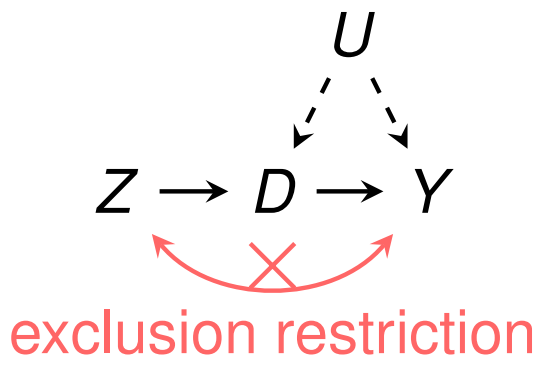
- Without U, we have error term $\gamma U_i + \eta_i$, and $\text{Cov}(\gamma U_i + \eta_i, D_i) \neq 0$.

The role of the instrument



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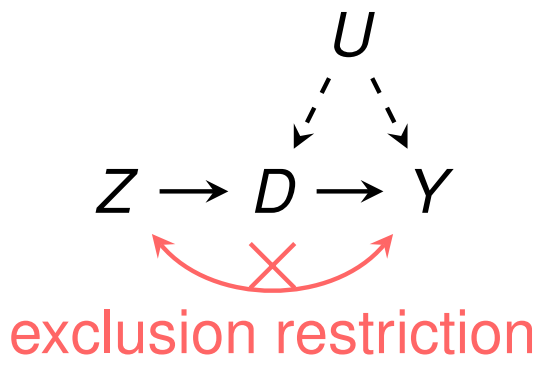

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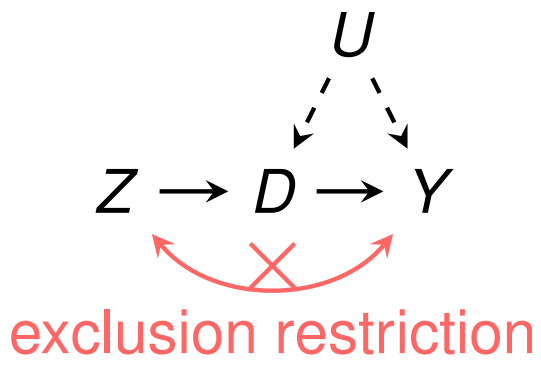

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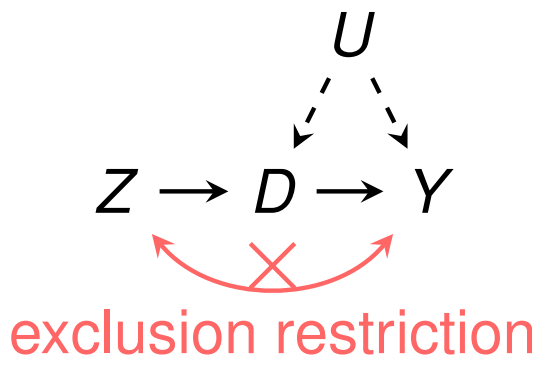

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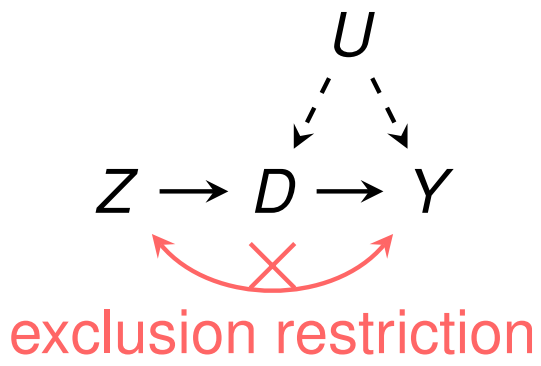

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&= 0 + \tau \text{Cov}(D_i, Z_i) + 0 \\
\implies \tau &= \frac{\text{Cov}(Y_i, Z_i)}{\text{Cov}(D_i, Z_i)}
\end{aligned}$$

IV estimator with constant effects

We can see this as a ratio of two linear regression estimates: treatment effect:

$$\tau = \frac{\text{Cov}(Y_i, Z_i)}{\text{Cov}(D_i, Z_i)} = \frac{\text{Cov}(Y_i, Z_i)/\mathbb{V}[Z_i]}{\text{Cov}(D_i, Z_i)/\mathbb{V}[Z_i]}$$

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So many words and names:

$$\begin{aligned} Wald = 2SLS = ILS &= \frac{\text{effect of } Z \text{ on } Y}{\text{effect of } Z \text{ on } D} \\ &= \frac{\text{Reduced form effect (aka intent-to-treat)}}{\text{First stage effect}} \end{aligned}$$

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More generically,

$$\tau_{IV} = (Z^\top X)^{-1}(Z^\top Y)$$

where X contains D and any covariates.

Assumptions

Let's take stock of the assumptions we have used:

1. Exclusion and exogeneity of instrument
2. Relevance
3. Constant treatment effects

Weak instruments

$$\widehat{\tau}_{IV} = \frac{\widehat{\mathrm{Cov}}(Y_i, Z_i)}{\widehat{\mathrm{Cov}}(D_i, Z_i)} \xrightarrow{p} \tau + \frac{\mathrm{Cov}(Z_i, U_i)}{\mathrm{Cov}(Z_i, D_i)}$$

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- Can easily be much worse than bias of OLS!
- Standard to communicate the strength of the first-stage. $F > 10$ is usual rule, but its just a rule.
- Better: Use Anderson-Rubin approach, which constructs CIs that have correct coverage even with weak instruments

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- ...but D_i is (still) **endogenous**: $\mathbb{E}[D_i \varepsilon_i] \neq 0$

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&= X_i^\top \beta + \tau[X_i^\top \alpha + \gamma Z_i] + [\tau \nu_i + \varepsilon_i] \\
&= X_i^\top \beta + \tau[X_i^\top \alpha + \gamma Z_i] + \varepsilon_i^* \\
&= X_i^\top \beta + \tau \mathbb{E}[D_i | X_i, Z_i] + \varepsilon_i^*
\end{aligned}
$$

- Because Z_i and X_i are uncorrelated with ν_i and ε_i, then $\mathbb{E}[D_i | X_i, Z_i]$ is also independent of ε_i^*.

Getting the reduced form

- We can plug the treatment equation into the outcome equation:

$$
\begin{aligned}
Y_i &= X_i^\top \beta + \tau[X_i^\top \alpha + \gamma Z_i + \nu_i] + \varepsilon_i \\
&= X_i^\top \beta + \tau[X_i^\top \alpha + \gamma Z_i] + [\tau \nu_i + \varepsilon_i] \\
&= X_i^\top \beta + \tau[X_i^\top \alpha + \gamma Z_i] + \varepsilon_i^* \\
&= X_i^\top \beta + \tau \mathbb{E}[D_i | X_i, Z_i] + \varepsilon_i^*
\end{aligned}
$$

- Because Z_i and X_i are uncorrelated with ν_i and ε_i, then $\mathbb{E}[D_i | X_i, Z_i]$ is also independent of ε_i^*.
- Thus, the population regression coefficient of a Y_i on $[X_i^\top \alpha + \gamma Z_i]$ is the average treatment effect, τ.

Two-stage least squares (2SLS)

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1. Run regression of treatment on covariates and instrument

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- This is not how we actually estimate 2SLS because the standard errors from second stage are wrong.

Two-stage least squares (2SLS)

Heuristic procedure:

1. Run regression of treatment on covariates and instrument
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- This is not how we actually estimate 2SLS because the standard errors from second stage are wrong.
- You can instead find that the 2SLS estimator is simply $(\mathbf{Z}^\top \mathbf{X})^{-1}(\mathbf{Z}^\top Y)$, where $\mathbf{Z} = [X, Z]$, and $\mathbf{X} = [X, D]$. Can compute SEs from there.